Advanced Signal Processing
ES7, ES9
Fall 2024

Linear Predictive Coding
- An Overview

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PK

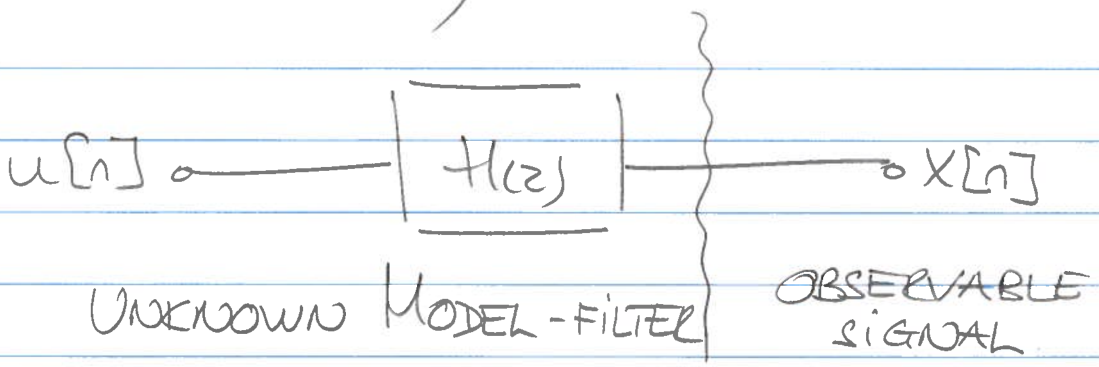
ADVANCED SIGNAL PROCESSING, ES-7

LINEAR PREDICTIVE CODING, I.

OUR PURPOSE IS TO DERIVE AN ALTERNATIVE TO THE TRADITIONAL DFT-BASED NON-PARAMETRIC SPECTRAL ANALYSIS.

WE ARE LOOKING FOR A MODEL-BASED, OR PARAMETRIC SIGNAL/SPECTRAL ANALYSIS METHOD.

THE FUNDAMENTAL IDEA IS TO ASSUME THAT THE OBSERVABLE SIGNAL $x[n]$ IS GENERATED BY AN LTI SYSTEM $H(z)$, EXCIDED BY AN INPUT SIGNAL $u[n]$;



IN THE GENERAL CASE, THE MODEL-FILTER CAN BE EXPRESSED AS

$$H(z) = \frac{X(z)}{U(z)} = G \cdot \frac{1 + \sum_{l=1}^{q} b_l \cdot z^{-l}}{1 + \sum_{k=1}^{p} a_k \cdot z^{-k}}$$

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$u[n] \longrightarrow \boxed{H(z)} \longrightarrow x[n]$

$$H(z) = \frac{X(z)}{U(z)} = G \cdot \frac{1 + \sum_{l=1}^{q} b_l \cdot z^{-l}}{1 + \sum_{k=1}^{p} a_k z^{-k}}$$

GAIN FACTOR.

THIS IS A GENERAL ARMA - MODEL

MA : MOVING AVERAGE (NUMERATOR)
AR : AUTO REGRESSIVE (DENOMINATOR)

IF WE MAKE THE ASSUMPTION THAT $x[n]$ IS THE RESPONSE OF $H(z)$ EXCIDED BY A WHITE SIGNAL $u[n]$, THEN $x[n]$ CONTAINS ALL INFORMATION ABOUT $H(z)$, AND THUS $|H(e^{j\omega})|$ REPRESENTS THE SPECTRAL CONTENT OF $x[n]$

THEREFORE, IF FOR GIVEN q AND p WE CAN ESTIMATE b_l, a_k AND G FROM $x[n]$, THEN WE HAVE CONDUCTED A MODEL-BASED SPECTRAL ANALYSIS OF $x[n]$.

CONSIDERING b_l, a_k AND G AS PARAMETERS, WE ALSO DENOTE THE METHOD AS

PARAMETRIC SPECTRAL ANALYSIS

LET'S NOW CONSIDER THE SITUATION WHERE WE USE ONLY THE AR-PART OF $H(z)$;

$$H(z) = \frac{X(z)}{U(z)} = \frac{G}{1 + \sum_{k=1}^{p} a_k z^{-k}}$$

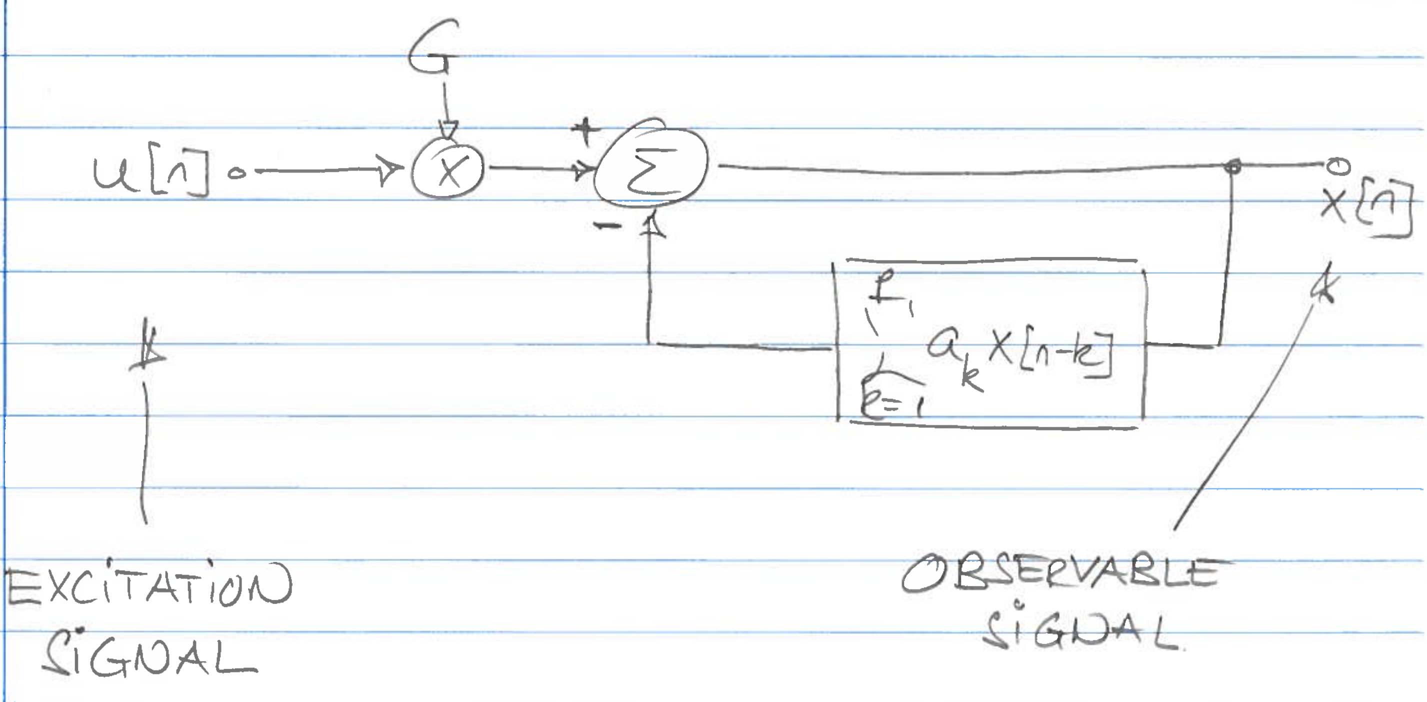
THE MODEL IS NOW AN AR-MODEL WITH MODEL-ORDER p. FIND a_k AND G!

$$X(z)\left\{1 + \sum_{k=1}^{p} a_k z^{-k}\right\} = G \cdot U(z)$$

$$\Big\Downarrow z^{-1}$$

$$x[n] = -\sum_{k=1}^{p} a_k x[n-k] + G \cdot u[n]$$

BLOCK DIAGRAM



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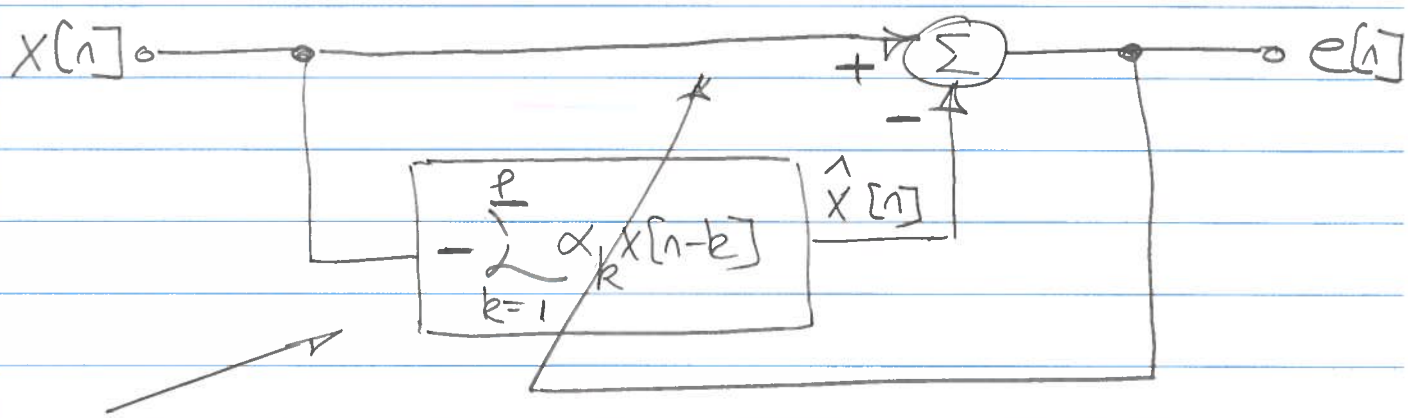
THE IDEA NOW IS THAT THE OBSERVABLE SIGNAL $x[n]$ CAN BE PREDICTED BASED ON PREVIOUSLY OBSERVED SAMPLE, I.E., $x[n-k]$ WHERE $k \geq 1$

WE THEREFORE INTRODUCE A ONE-STEP PREDICTOR OF ORDER p



LINEAR p^{th} ORDER
1-STEP PREDICTOR

$\hat{x}[n]$ IS A PREDICTED VALUE OF $x[n]$, AND THE PREDICTION ERROR $e[n] = x[n] - \hat{x}[n]$

$\boxed{\text{NOW, ADJUST THE PARAMETER } \alpha_k \text{ SUCH THAT } e[n] \text{ IS MINIMIZED „IN SOME SENSE"}}$

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$$e[n] = x[n] - \hat{x}[n]$$

$\Downarrow$

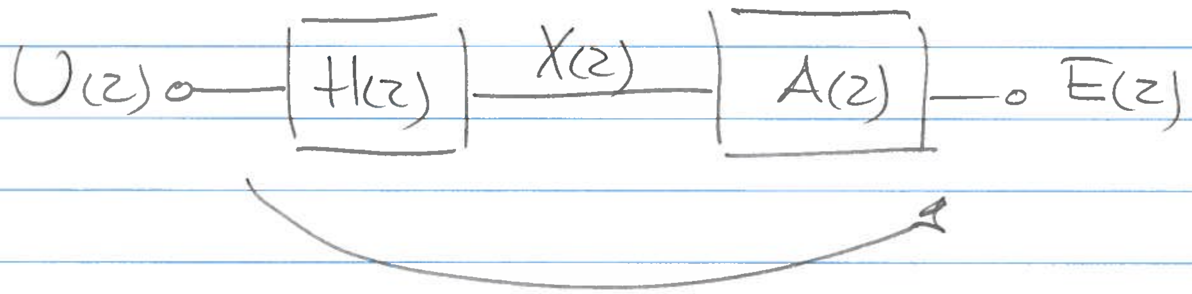
$$e[n] = x[n] + \sum_{k=1}^{p} \alpha_k x[n-k]$$

$Z \Downarrow$

$$E(z) = X(z) + \sum_{k=1}^{p} \alpha_k X(z) z^{-k}$$

$\Downarrow$

$$\frac{E(z)}{X(z)} = A(z) = \underbrace{1 + \sum_{k=1}^{p} \alpha_k z^{-k}}_{\text{FIR FILTER}}$$



THE TOTAL TRANSFER FUNCTION

$$\frac{E(z)}{U(z)} = H(z) \cdot A(z) = \frac{G}{1 + \sum_{k=1}^{p} a_k z^{-k}} \cdot \left\{ 1 + \sum_{k=1}^{p} \alpha_k z^{-k} \right\}$$

$$= G \quad \text{for } a_k = \alpha_k,\ k = 1..p$$

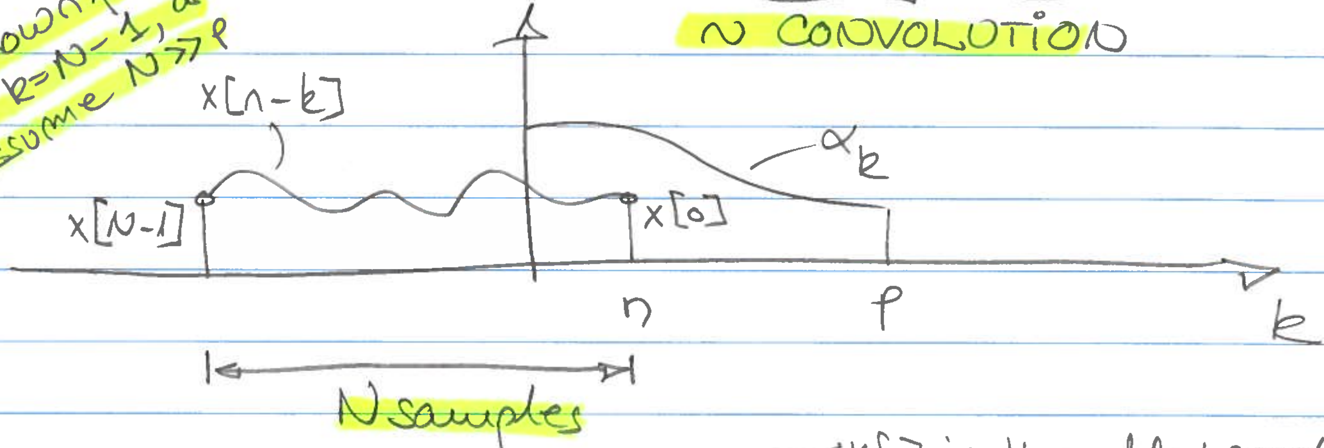
$\Downarrow$

$$\boxed{A(z) = \frac{G}{H(z)} \quad \text{THE INVERSE FILTER}}$$

A MATRIX FORMULATION

$$e[n] = x[n] + \sum_{k=1}^{p} \alpha_k x[n-k] = \underbrace{\sum_{k=0}^{p} \alpha_k x[n-k]}_{\sim \text{ CONVOLUTION}}, \quad \alpha_0 = 1.$$

$x[k]$ is known from $k=0$ to $k=N-1$, and we assume $N \gg p$



x[0] is the oldest sample

$$e[0] = \alpha_0 x[0]$$
$$e[1] = \alpha_0 x[1] + \alpha_1 x[0]$$
$$e[2] = \alpha_0 x[2] + \alpha_1 x[1] + \alpha_2 x[0]$$

$\vdots$

$$e[p] = \alpha_0 x[p] + \quad \cdots \quad + \alpha_p x[0]$$

$\vdots \qquad\qquad\qquad\qquad \vdots$

$$e[N-1] = \alpha_0 x[N-1] + \quad \cdots \quad + \alpha_p x[N-1-p]$$

$\vdots \qquad\qquad\qquad\qquad \vdots$

$$e[N-1+p] = \qquad\qquad\qquad \alpha_p x[N-1]$$

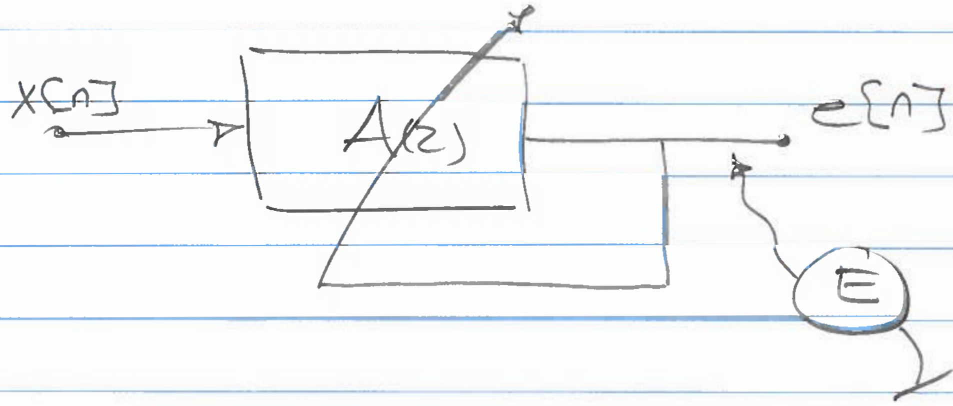
$$\left[\begin{array}{c|ccccc} x_0 & 0 & 0 & \cdots & & 0 \\ x_1 & x_0 & 0 & & & \vdots \\ x_2 & x_1 & x_0 & & & \vdots \\ \vdots & \vdots & & \ddots & & 0 \\ x_p & x_{p-1} & & \cdots & & x_0 \\ \vdots & \vdots & & & & \vdots \\ x_{N-1} & x_{N-2} & & & & x_{N-1-p} \\ 0 & 0 & & & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 0 & x_{N-1} \end{array}\right] \left[\begin{array}{c} [1] \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{array}\right] = \left[\begin{array}{c} e_0 \\ e_1 \\ \vdots \\ e_p \\ \vdots \\ e_{N-1} \\ \vdots \\ e_{N-1+p} \end{array}\right] \Bigg\updownarrow N+p$$

p

$-\underline{b}$ $\quad \underline{\underline{X}}$ $\quad \underline{a}$ $\quad \underline{e}$

$$\boxed{\underline{\underline{X}}\,\underline{a} - \underline{b} = \underline{e}}$$

AN IDEA NOW IS TO MINIMIZE THE ENERGY IN THE PREDICTION ERROR $e[n]$



E IS KNOWN AS THE ERROR ENERGY OR THE ERROR VARIANCE

$$E = \sum_{n=-\infty}^{\infty} e^2[n] = \sum_{n=0}^{N-1+p} e^2[n] = \underline{e}^T \cdot \underline{e}$$

$$= (\underline{\underline{X}}\,\underline{a} - \underline{b})^T (\underline{\underline{X}}\,\underline{a} - \underline{b})$$

Definition of e from p. 7

$$= (\underline{a}^T \underline{\underline{X}}^T - \underline{b}^T)(\underline{\underline{X}}\,\underline{a} - \underline{b})$$

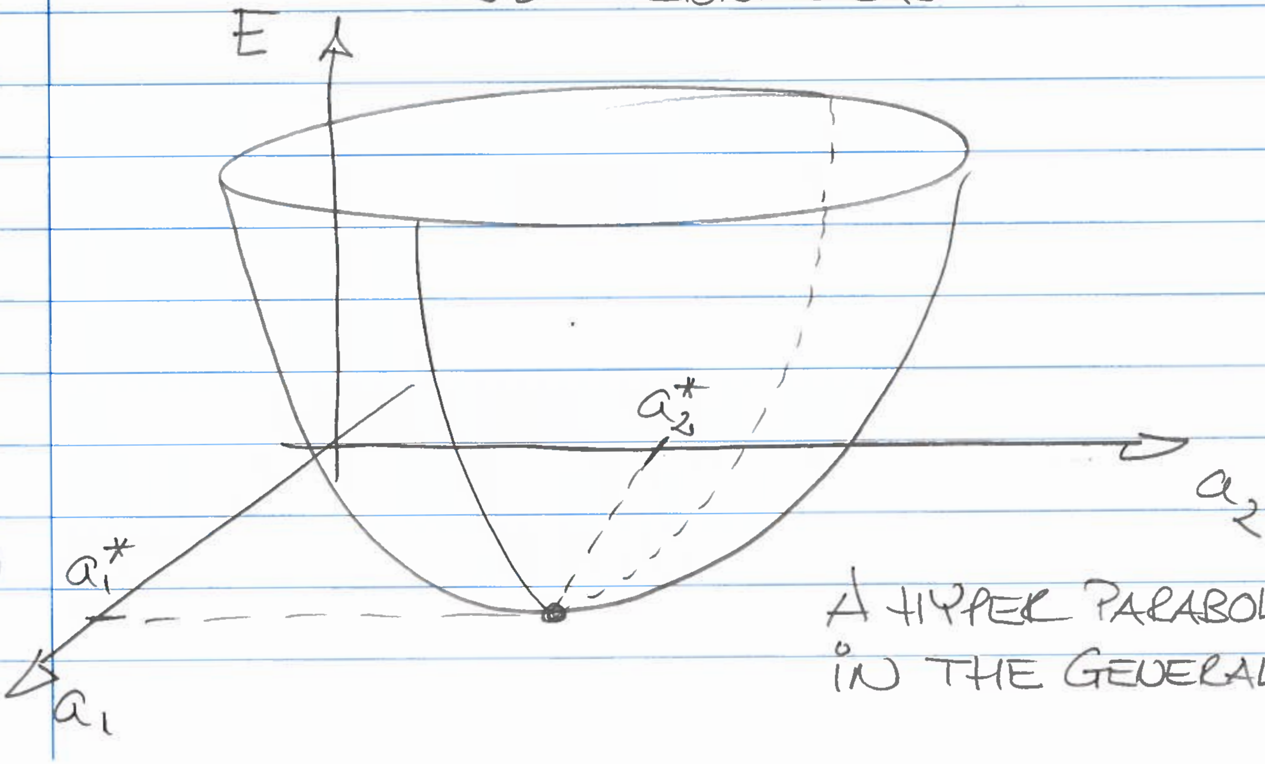
$$= \underline{a}^T \underline{\underline{X}}^T \underline{\underline{X}}\,\underline{a} - \underbrace{\underline{a}^T \underline{\underline{X}}^T \underline{b}} - \underbrace{\underline{b}^T \underline{\underline{X}}\,\underline{a}} + \underline{b}^T \underline{b}$$

TRANSPOSED OF EACH OTHER (SCALAR)

$$= \underline{a}^T \underline{\underline{X}}^T \underline{\underline{X}}\,\underline{a} - 2\,\underline{a}^T \underline{\underline{X}}^T \underline{b} + \underline{b}^T \underline{b}$$

SO, WHAT WE SEE HERE IS THAT E IS A QUADRATIC FUNCTION OF $\underline{a}$

2 DIMENSIONAL CASE



A HYPER PARABOLOID IN THE GENERAL CASE.

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FIND THE MINIMAL VALUE OF E

1ST METHOD: DIFFERENTIATE E W.R.T. $\underline{a}$ AND SET EQUAL TO $\underline{0}$

$$\frac{\partial E}{\partial \underline{a}^T} = \underline{0}$$

WE USE $\underline{a}^T$ BECAUSE IT LEADS TO A COLUMN VECTOR ON THE RIGHT HAND SIDE.

$$\Downarrow$$

$$\frac{\partial}{\partial \underline{a}^T}\left\{\underline{a}^T \underline{\underline{X}}^T \underline{\underline{X}}\,\underline{a} - 2\underline{a}^T \underline{\underline{X}}^T \underline{b} + \underline{b}^T \underline{b}\right\} = \underline{0}$$

$$\Downarrow$$

$$2 \cdot \underline{\underline{X}}^T \underline{\underline{X}}\,\underline{a} - 2\underline{\underline{X}}^T \underline{b} = \underline{0}$$

$$\Downarrow$$

$$\boxed{\underline{\underline{X}}^T \underline{\underline{X}}\,\underline{a} = \underline{\underline{X}}^T \underline{b}}$$

NORMAL EQUATIONS

$$\Downarrow$$

$$\underline{a}_{OPT} = \underbrace{\left(\underline{\underline{X}}^T \underline{\underline{X}}\right)^{-1} \underline{\underline{X}}^T}_{\underline{\underline{X}}^{\#}} \underline{b}$$

IF THE INVERSE EXISTS...

THE PSEUDO INVERSE

$$\Downarrow$$

$$\boxed{\underline{a}_{OPT} = \underline{\underline{X}}^{\#} \underline{b}}$$

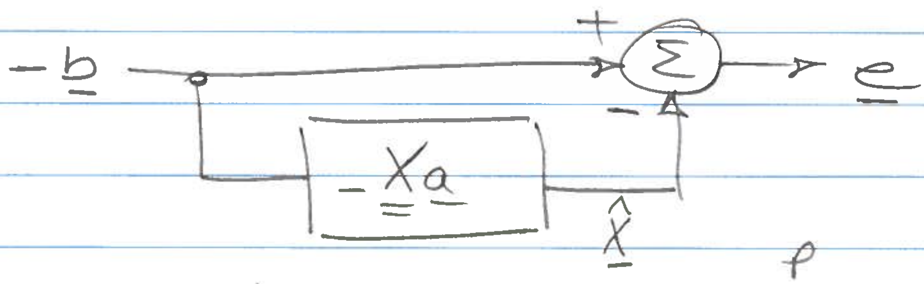
2^{ND} METHOD : USING THE ORTHOGONALITY PRINCIPLE

- THE MATRIX $\underline{\underline{X}}$ CAN BE WRITTEN AS ;

$$\underline{\underline{X}} = \{ \underline{x}_1 \ \underline{x}_2 \ \underline{x}_3 \cdots \underline{x}_p \}$$

FROM WHICH WE SEE THAT IT SPANS THE VECTOR SPACE $\mathbb{R}^j$, $1 \le j \le p$

- THE LINEAR PREDICTOR EXPRESSED IN MATRIX NOTATION ;

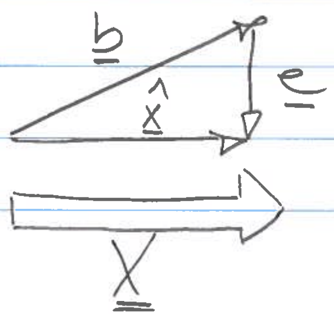


SO, $\hat{\underline{x}} = -\underline{\underline{X}}\,\underline{a} = \{ \underline{x}_1 \underline{x}_2 \underline{x}_3 \cdots \underline{x}_p \} \underline{a} = \sum_{k=1}^{p} a_k \underline{x}_k$

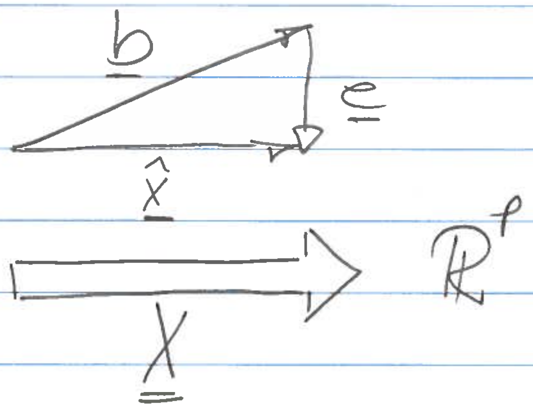
AND THEREFORE $\hat{\underline{x}} \in \mathbb{R}^p$

- $\underline{\underline{X}}\,\underline{a} = -\hat{\underline{x}}$ AND $\underline{e} = -\underline{b} - (-\hat{\underline{x}})$

$$\Downarrow \quad \hat{\underline{x}} = \underline{b} + \underline{e}$$



VECTOR SPACE $\mathbb{R}^p$



FROM THIS ILLUSTRATION IT IS CLEAR, THAT $E = e^T e = |e|^2$ IS MINIMAL WHEN

$$e \perp x_i \quad \forall i \qquad 1 \leq i \leq p$$

$$\Downarrow$$

$$\boxed{X^T \cdot e = 0}$$

FROM THE MATRIX DEFINITION ON P.7 WE HAVE,

$$Xa = b + e$$

$$\Downarrow$$

$$X^T X a = X^T b + \cancel{X^T \cdot e} \to 0$$

$$\Downarrow$$

$$a_{OPT} = (X^T X)^{-1} X^T b$$

$$\Downarrow$$

$$\boxed{a_{OPT} = X^{\#} b}$$

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INTERPRETATION OF NORMAL EQU.

From p. 9

$$\underline{\underline{X}}^T\underline{\underline{X}} = \begin{bmatrix} \sum_{i=0}^{N-1} x^2[i] & \sum_{i=0}^{N-1} x[i+1]\cdot x[i] & \cdots \\ \sum_{i=0}^{N-1} x[i]\cdot x[i-1] & \ddots & \\ \vdots & & \ddots \end{bmatrix}$$

$$= \begin{bmatrix} R(0) & R(1) & R(2) & \cdots & R(p-1) \\ R(1) & R(0) & R(1) & \cdots & R(p-2) \\ \vdots & & \ddots & & \vdots \\ R(p-1) & R(p-2) & \cdots & & R(0) \end{bmatrix}$$

$$= \underline{\underline{R}}$$ SYMMETRIC AND TOEPLITZ AUTOCORRELATION MATRIX

$$\underline{\underline{X}}^T\underline{b} = \begin{bmatrix} -r(1) \\ -r(2) \\ \vdots \\ -r(p) \end{bmatrix} = -\underline{r}$$ AUTOCORR. VECTOR.

$$\underline{\underline{X}}^T\underline{\underline{X}}\,\underline{a} = \underline{\underline{X}}^T\cdot\underline{b} \Rightarrow \boxed{\underline{\underline{R}}\,\underline{a} = -\underline{r}}$$

YULE-WALKER EQUATIONS

• EXPRESSION FOR MINIMAL ERROR-VARIANCE

— ALSO KNOWN AS THE RESIDUAL-VARIANCE

$E = \underline{e}^T \underline{e}$ is minimal for $\underline{a} = \underline{a}_{opt} = (\underline{\underline{X}}^T \underline{\underline{X}})^{-1} \underline{\underline{X}}^T \underline{b}$ p.9 and p.11

$E = \underline{a}^T \underline{\underline{X}}^T \underline{\underline{X}}\, \underline{a} - 2\underline{a}^T \underline{\underline{X}}^T \underline{b} + \underline{b}^T \underline{b}$ p.8

$\Downarrow$

$$E_{min} = \underline{a}_{opt}^T \underline{\underline{X}}^T \underline{\underline{X}}\, \underline{a}_{opt} - 2\underline{a}_{opt}^T \underline{\underline{X}}^T \underline{b} + \underline{b}^T \underline{b}$$

$\Downarrow$

$$E_{min} = \underline{b}^T \underline{\underline{X}} (\underline{\underline{X}}^T \underline{\underline{X}})^{-1} \cancel{\underline{\underline{X}}^T \underline{\underline{X}}\, (\underline{\underline{X}}^T \underline{\underline{X}})^{-1}} \underline{\underline{X}}^T \underline{b}$$
$$- 2\underline{b}^T \underline{\underline{X}} (\underline{\underline{X}}^T \underline{\underline{X}})^{-1} \underline{\underline{X}}^T \underline{b} + \underline{b}^T \underline{b}$$

$\Downarrow$

$$E_{min} = \underline{b}^T \underline{b} - \underbrace{\underline{b}^T \underline{\underline{X}}}_{\text{row vector}} \underbrace{(\underline{\underline{X}}^T \underline{\underline{X}})^{-1} \underline{\underline{X}}^T \underline{b}}_{\underline{a}_{opt} \text{ (column vector)}}$$

$\Downarrow$

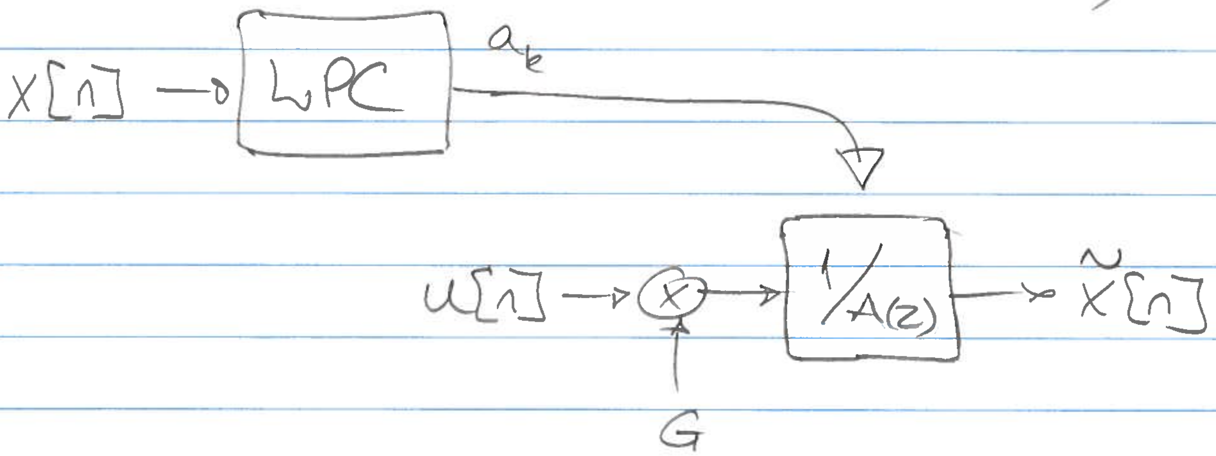
$$\boxed{E_{min} = R(0) + \sum_{k=1}^{p} a_k \cdot R(k)}$$

SO, THE MINIMAL RESIDUAL-VARIANCE IS A LINEAR COMBINATION OF ALL p AUTOCORRELATION LAGS.

Discussion on the Gain, G.

Let's assume the scenario where we have estimated all the model parameters a_k based on the observable signal $x[n]$.

We now want to use these parameters to generate an estimate $\tilde{x}[n]$ of $x[n]$;



In this context, we need to calculate G.

One approach to do this is to require that the energy in the synthesized signal $\tilde{x}[n]$ is identical to the energy in the observable signal $x[n]$;

$$E\left[\tilde{x}^2[n]\right] = E\left[x^2[n]\right]$$

Let's assume that the excitation signal $u[n]$ is a white signal;

$$E\left[u[n] \cdot u[n-i]\right] = \delta[i]$$

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$$u[n] \longrightarrow \overset{G}{\otimes} \longrightarrow \boxed{1/A(z)} \longrightarrow \tilde{x}[n]$$

$$\tilde{x}[n] = -\sum_{k=1}^{p} a_k \cdot \tilde{x}[n-k] + G \cdot u[n] \qquad (*)$$

$$\Downarrow$$

$$\tilde{x}[n] \cdot \tilde{x}[n-i] = -\sum_{k=1}^{p} a_k \tilde{x}[n-k] \cdot \tilde{x}[n-i] + G u[n] \cdot \tilde{x}[n-i]$$

$$\Downarrow$$

$$0 \leq i \leq p$$

$$E\{\tilde{x}[n] \cdot \tilde{x}[n-i]\} = -\sum_{k=1}^{p} a_k E\{\tilde{x}[n-k] \cdot \tilde{x}[n-i]\}$$

$$+ G \cdot E\{u[n] \cdot \tilde{x}[n-i]\} \qquad 0 \leq i \leq p$$

WE NOW EVALUATE THIS EXPRESSION FOR TWO CASES;

- $1 \leq i \leq p$

$$\tilde{R}(i) = -\sum_{k=1}^{p} a_k \cdot \tilde{R}(k-i) + G \cdot 0$$

- $i = 0$

$$\tilde{R}(0) = -\sum_{k=1}^{p} a_k \cdot \tilde{R}(k) + G \cdot E\{u[n] \cdot \tilde{x}[n]\}$$

INSERT $(*)$

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$$G \cdot E\left\{u[n] \cdot \left(-\sum_{k=1}^{p_1} a_k \cdot \tilde{x}[n-k] + G \cdot u[n]\right)\right\}$$

SINCE $u[n]$ IS WHITE, THERE IS NO CORRELATION BETWEEN $u[n]$ AND DELAYED SAMPLES OF $\tilde{x}[n]$, i.E., $\tilde{x}[n-k]$. THEREFORE;

$$G \cdot E\left\{-\sum_{k=1}^{p_1} a_k u[n] \cdot \tilde{x}[n-k] + G \cdot u^2[n]\right\}$$

$$= G \cdot E\{G \cdot u^2[n]\} = G^2 \cdot \delta[0] = G^2$$ AND THUS;

$$\tilde{R}(0) = -\sum_{k=1}^{p_1} a_k \cdot \tilde{R}(k) + G^2$$

AND OUR REQUIREMENT WAS $\tilde{R}(0) = R(0)$

NOW, FROM THE CASE $1 \leq i \leq p$ WE HAVE;

$$\tilde{R}(i) = -\sum_{k=1}^{p_1} a_k \tilde{R}(k-i)$$

Yule-Walker equ., p.12

IF WE COMPARE THIS AGAINST $\underline{r} = -\underline{\underline{R}}\,\underline{a}$

WE CONCLUDE $\tilde{R}(k) = R(k) \quad 1 \leq k \leq p$

AND THUS;

$$\tilde{R}(0) = R(0) = -\sum_{k=1}^{p_1} a_k R(k) + G^2$$

$\Downarrow$

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$$G^2 = R(0) + \sum_{k=1}^{p} a_k R(k) = E_{min}$$ c.f. p. 13

$$\Downarrow$$

$$G = \sqrt{E_{min}}$$ for given model order p

EFFICIENT CALCULATION OF $\underline{\underline{R}} \cdot \underline{a} = -\underline{r}$

— THE LEVINSON-DURBIN ALGORITHM

THE IDEA IS TO CALCULATE THE LPC-COEFFICIENTS a_k RECURSIVELY, I.E., THE m^{th} ORDER LPC COEFFICIENTS ARE CALCULATED BASED ON THE $(m-1)^{st}$ ORDER COEFFICIENTS.

$$\underline{a}_m = \begin{bmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mm} \end{bmatrix} = \begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + \begin{bmatrix} \underline{d}_{m-1} \\ k_m \end{bmatrix}$$

KNOWING $\underline{a}_{m-1}$ OUR JOB NOW IS TO DETERMINE $\underline{d}_{m-1}$ AND k_m

WE START BY RE-ORGANIZING $\underline{\underline{R}}_m$

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$$\underline{\underline{R}}_m = \begin{bmatrix} \underline{\underline{R}}_{m-1} & \underline{r}^r_{m-1} \\ \underline{r}^{rT}_{m-1} & r(0) \end{bmatrix}$$

WHERE $\underline{r}^r_{m-1} = \{ r(m-1) \; r(m-2) \; \cdots r(1) \}^T$

IS THE VECTOR $\underline{r}_{m-1}$ WITH ITS ELEMENTS IN REVERSE ORDER (SYMMETRIC REFLECTION OF $\underline{r}_{m-1}$)

THE m^{th} ORDER SYSTEM IS DENOTED;

$$\underline{\underline{R}}_m \cdot \underline{a}_m = -\underline{r}_m$$

$\Downarrow$

$$\begin{bmatrix} \underline{\underline{R}}_{m-1} & \underline{r}^r_{m-1} \\ \underline{r}^{rT}_{m-1} & r(0) \end{bmatrix} \left[\begin{Bmatrix} \underline{a}_{m-1} \\ 0 \end{Bmatrix} + \begin{Bmatrix} \underline{d}_{m-1} \\ k_m \end{Bmatrix} \right] = -\begin{bmatrix} \underline{r}_{m-1} \\ r(m) \end{bmatrix}$$

- UPPER ROW;

$$\underbrace{\underline{\underline{R}}_{m-1} \underline{a}_{m-1}}_{-\underline{r}_{m-1}} + \underline{\underline{R}}_{m-1} \underline{d}_{m-1} + k_m \underline{r}^r_{m-1} = -\underline{r}_{m-1}$$

$\Downarrow$

$$\underline{\underline{R}}_{m-1} \underline{d}_{m-1} + k_m \cdot \underline{r}^r_{m-1} = \underline{0}$$

$\Downarrow$

$$\underline{d}_{m-1} = -k_m \cdot \underline{\underline{R}}^{-1}_{m-1} \cdot \underline{r}^r_{m-1}$$

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$$\underbrace{\underline{d}_{m-1} = -k_m \cdot \underline{\underline{R}}_{m-1}^{-1} \cdot \underline{r}^{r}_{m-1}}$$

LOOKS VERY MUCH LIKE THE SOLUTION TO THE $(m-1)^{st}$ ORDER SYSTEM, EXCEPT FOR THE REVERSED ELEMENT AUTO-CORRELATION VECTOR $\underline{r}^{r}_{m-1}$ AND THE k_m SCALAR.

DUE TO THE SYMMETRIC TOEPLITZ PROPERTY OF THE SYSTEM MATRIX, WE CAN WRITE

$$\underline{\underline{R}}_{m-1} \cdot \underline{a}^{r}_{m-1} = -\underline{r}^{r}_{m-1}$$

AND THUS; $\underline{d}_{m-1} = k_m \cdot \underline{a}^{r}_{m-1}$ (*)

- LOWER ROW

$$\underline{r}^{rT}_{m-1} \cdot \underline{a}_{m-1} + \underline{r}^{rT}_{m-1} \cdot \underline{d}_{m-1} + k_m \cdot r(0) = -r(m)$$

WE NOW INSERT THE UPDATE VECTOR $\underline{d}_{m-1}$, (*)

$$\underline{r}^{rT}_{m-1} \cdot \underline{a}_{m-1} + \underline{r}^{rT}_{m-1}\left(k_m \cdot \underline{a}^{r}_{m-1}\right) + k_m \cdot r(0) = -r(m)$$

$$\Downarrow$$

$$k_m\left(r(0) + \underline{r}^{rT}_{m-1} \cdot \underline{a}^{r}_{m-1}\right) = -r(m) - \underline{r}^{rT}_{m-1} \cdot \underline{a}_{m-1}$$

$$\Downarrow$$

$$k_m = -\frac{r(m) + \underline{r}^{rT}_{m-1} \cdot \underline{a}_{m-1}}{r(0) + \underline{r}^{rT}_{m-1}\, \underline{a}^{r}_{m-1}}$$

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$$k_m = -\frac{r(m) + \underline{r}^{r\,T}_{m-1} \cdot \underline{a}_{m-1}}{r(0) + \underline{r}^{r\,T}_{m-1} \cdot \underline{a}^{r}_{m-1}}$$

FOR AN m^{th} ORDER SYSTEM, WE HAVE (p. 13)

$$E_m = r(0) + \sum_{k=1}^{m} a_{mk} \cdot r(k)$$

AND THEREFORE WE CONCLUDE THAT THE DENOMINATOR EQUALS THE RESIDUAL VARIANCE OF THE $(m-1)^{st}$ ORDER PREDICTOR;

$$k_m = -\frac{r(m) + \underline{r}^{r\,T}_{m-1} \cdot \underline{a}_{m-1}}{E_{m-1}}$$

THESE VALUES, k_m, ARE APPLIED HERE MOSTLY AS „DUMMY VARIABLE", BUT IT TURNS OUT THAT THEY ARE VERY ESSENTIAL AND USEFUL COEFFICIENTS KNOWN AS THE REFLECTION OR PARCOR COEFFICIENTS

$$\underline{a}_m = \begin{bmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mm} \end{bmatrix} = \begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + \begin{bmatrix} \underline{d}_{m-1} \\ k_m \end{bmatrix} = \begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + \begin{bmatrix} k_m \cdot \underline{a}^{r}_{m-1} \\ k_m \end{bmatrix}$$

c.f. (*) p. 19

$$= \begin{bmatrix} \underline{a}_{m-1} \\ 0 \end{bmatrix} + k_m \begin{bmatrix} \underline{a}^{r}_{m-1} \\ 1 \end{bmatrix}$$

$\Downarrow$

PK

(Δ)

$$a_{m,k} = a_{m-1,k} + k_m \cdot a_{m-1,m-k} \qquad \begin{cases} m = 1,2,\dots p \\ k = 1,2,\dots m- \end{cases}$$

$$a_{m,m} = k_m$$

WHERE m DENOTES THE ACTUAL ORDER, AND k (not to be confused with the reflection coefficients) IS A VARIABLE USED FOR NUMBERING THE LPC COEF.

FROM p.20 WE SEE THAT THE k_m USED FOR UPDATING THE LPC VECTOR $\underline{a}$, IS BASED ON THE $(m-1)^{st}$ ORDER RESIDUAL VARIANCE;

$$E_m = r(0) + \sum_{k=1}^{m} a_{mk} \cdot r(k)$$

WE NOW SUBSTITUTE (Δ)

$$E_m = r(0) + \sum_{k=1}^{m} \{a_{m-1,k} + k_m \cdot a_{m-1,m-k}\} \cdot r(k)$$

$\Downarrow$

$$E_m = r(0) + \sum_{k=1}^{m} a_{m-1,k} \cdot r(k) + \sum_{k=1}^{m} a_{mm} \cdot a_{m-1,m-k} \cdot r(k)$$

$\Downarrow$

$$E_m = r(0) + \sum_{k=1}^{m-1} a_{m-1,k} \cdot r(k) + a_{mm} \left\{ r(m) + \sum_{k=1}^{m-1} a_{m-1,m-k} \cdot r(k) \right\}$$

DECREASING THE UPPER BOUND FROM m TO $m-1$ IS POSSIBLE SINCE ONLY $m-1$ LPC COEF. ARE USED IN THE $(m-1)^{st}$ ORDER PREDICTOR.

PK

$$E_m = r(0) + \sum_{k=1}^{m-1} a_{m-1,k} \cdot r(k) + a_{mm} \underbrace{\left\{ r(m) + \sum_{k=1}^{m-1} a_{m-1,m-k} \cdot r(k) \right\}}$$

EQUALS THE NUMERATOR IN k_m p. 20;

$$k_m = - \frac{r(m) + \underline{r}_{m-1}^{r\,T} \cdot \underline{a}_{m-1}}{E_{m-1}}$$

$$\Downarrow$$

$$k_m \cdot E_{m-1} = -\left(r(m) + \underline{r}_{m-1}^{r\,T} \cdot \underline{a}_{m-1} \right)$$

SO, IT IS POSSIBLE TO CALCULATE THE RESIDUAL VARIANCE FOR THE m^{th} ORDER PREDICTOR BASED ON THE RESIDUAL VARIANCE FOR THE $(m-1)^{st}$ ORDER PREDICTOR;

$$E_m = \underbrace{r(0) + \left\{ \sum_{k=1}^{m-1} a_{m-1,k} \cdot r(k) \right\}}_{E_{m-1} \text{ c.f. p13}} - a_{mm} \cdot k_m \cdot E_{m-1}$$

a_{mm}: k_m c.f. p. 21

$$\Downarrow$$

$$\boxed{E_m = E_{m-1} \left(1 - k_m^2 \right)}$$

PK

THE FINAL LEVINSON-DURBIN ALGORITHM

$E_0 = r(0);$

for $m = 1$ to p

$$k_m = -\frac{r(m) + \sum_{k=1}^{m-1} a_{m-1,k} \cdot r(m-k)}{E_{m-1}};$$

Calculation of k, p.20

$a_{mm} = k_m;$

for $k = 1$ to $m-1$

$$a_{m,k} = a_{m-1,k} + k_m \cdot a_{m-1,m-k};$$

end

Calculation of a, p. 21

$$E_m = E_{m-1} \cdot (1 - k_m^2);$$

end

Calculation of E, p.22

Final Remarks:

Using the Levinson-Durbin algorithm for solving a p^{th} order system, all LPC coefficients for lower order systems are also calculated

$$\begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{p1} \\ & a_{22} & a_{32} & \cdots & a_{p2} \\ & & a_{33} & \cdots & a_{p3} \\ & & & & \vdots \\ & & & & a_{pp} \end{bmatrix}$$

On the diagonal, we have all the reflections coefficients k_m.